

Paul Haskins

March 26, 2025

$$\vec{\nabla} \times \vec{H} - \frac{\partial \vec{D}}{\partial t} = \vec{j}_{free} \quad (1)$$

$$\vec{\nabla} \cdot \vec{D} = \rho_{free} \quad (2)$$

$$\vec{\nabla} \times \vec{E} + \frac{\partial \vec{B}}{\partial t} = 0, \quad (3)$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad (4)$$

Focus on dielectric media, i.e. electric insulators that can be polarised

$$\rho_{free} = 0, \quad \vec{j}_{free} = 0 \quad (5)$$

In addition we assume linear media

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} = \epsilon \vec{E} \quad (6)$$

$$\vec{H} = \frac{1}{\mu_0} - \vec{M} = \frac{1}{\mu} \vec{B} \quad (7)$$

Use these to replace $\vec{D}$ and $\vec{H}$

$$\frac{1}{\mu} \vec{\nabla} \times \vec{B} - \mu \epsilon \frac{\partial \vec{E}}{\partial t} = \vec{0}, \quad (8)$$

$$\epsilon \vec{\nabla} \cdot \vec{E} = 0 \quad (9)$$

$$\vec{\nabla} \times \vec{E} + \frac{\partial \vec{B}}{\partial t} = 0 \quad (10)$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad (11)$$

Differentiate eqn (8) wrt t

$$\implies \vec{\nabla} \times \frac{\partial \vec{B}}{\partial t} - \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2} = \vec{0} \quad (12)$$

$$\frac{\partial \vec{B}}{\partial t} = -\vec{\nabla} \times \vec{E} \quad (13)$$

$$-\vec{\nabla} \times (\gamma \times \vec{E}) - \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2} = 0 \quad (14)$$

$$-\gamma \times (\gamma \times \vec{E}) = -\vec{\nabla}^2 \vec{E} \text{ if } \vec{\nabla} \cdot \vec{E} = 0 \quad (15)$$

$$\mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2} - \vec{\nabla}^2 \cdot \vec{E} = 0 \quad (16)$$

We can do the same for $\vec{B}$ analogously we get

$$\mu \epsilon \frac{\partial^2 \vec{B}}{\partial t^2} - \vec{\nabla}^2 \cdot \vec{B} = 0 \quad (17)$$

These are the wave equation for the $\vec{E}$ and $\vec{B}$ fields velocity $v = \frac{1}{\sqrt{\mu \epsilon}}$. Here we have worked in XXX units while in the SI units we would easily identify the d Lambert operator.

Make an ansatz due to the fact that the $\vec{E}$ field is linear in our wave eqn

$$\vec{E}(\vec{x}, t) = \vec{E}_0 e^{i(\vec{k} \cdot \vec{x} - \omega t)} \quad (18)$$

These fields are complex, but physical fields must be real.

The physical $\vec{E}$, $\vec{B}$ fields correspond to $\Re \vec{E}$ and $\Re \vec{B}$.

In Griffith's $\vec{E}$ for complex fields $\vec{E}$ and $\vec{B}$ for physical fields.

Due to the linearity of the wave equation (and the MW eqs)

$$\Re \left[\left(\frac{1}{v^2} \frac{\partial^2}{\partial t^2} - \vec{\nabla}^2 \right) \vec{E}(\vec{x}, t) \right] = \left(\frac{1}{v^2} \frac{\partial^2}{\partial t^2} - \vec{\nabla}^2 \right) \Re \vec{E}(\vec{x}, t) \quad (19)$$

Complexified fields make the algebra easier. Observables are often the energy flux often quadratic in the fields.

WARNING

Quantities that are non-linear in the fields e.g.

$$\vec{S} \propto \vec{E} \times \vec{B} \equiv (\Re \vec{E}) \times (\Re \vec{B}) \neq \Re (\vec{E} \times \vec{B}) \quad (20)$$

$\exists$ a trick to get energy transport in plane wave (maybe on exam).

Now to use the ansatz

$$\frac{\partial \vec{E}}{\partial t} = -i\omega \vec{E}, \quad \frac{\partial \vec{E}}{\partial x_i} = ik_i \vec{E} \quad (21)$$

$$\frac{\partial^2}{\partial t^2} \vec{E} = (-i\omega)^2 \vec{E}, \quad \frac{\partial^2}{\partial x_i^2} \vec{E} = (ik_i)^2 \vec{E} \quad (22)$$

$$\left[\frac{1}{v^2} (-i\omega)^2 - (\vec{k})^2 \right] \vec{E} = 0 \quad (23)$$

$$= \frac{\omega^2}{v^2} + \vec{k}^2 = 0 \text{ or } \vec{k}^2 = \frac{\omega^2}{v^2} \quad (24)$$

Why a plane wave?

$$\vec{E}(\vec{x}, t) = \vec{E}_0 e^{i\phi(\vec{x}, t)} \quad (25)$$

phase $\phi(\vec{x}, t) = \vec{k} \cdot \vec{x} - \omega t$

A $t = t_0$ phase $\phi_0 = \phi(\vec{x}, t_0) = \vec{k} \cdot \vec{x} - \omega t_0$ keep fixed

Solutions $\vec{x}$ of $\phi(\vec{x}, t_0) = \phi_0$ all points for which $\vec{k} \cdot \vec{x} = \phi_0 + \omega t_0 = \text{const} = |\vec{k}| \cdot x_{||}$ (parallel to $\vec{k}$).

INCLUDE IDAGRMA XXX

$$\frac{\partial}{\partial t} \phi(\vec{x}, t) = |\vec{k}| \cdot \dot{x}_{||} - \omega = 0 \quad (26)$$

$$\implies \dot{x}_{||} = \frac{\omega}{|\vec{k}|} = v \quad (27)$$

$\implies$ Wave front moves with phase velocity in direction $\hat{k}$

Wavelength $e^{i\phi} = e^{i\phi + 2\pi m i}$ $m \in \mathbb{Z}$. if

$$(x_{||} + \Delta x_{||})k - \omega t = x_{||}k - \omega t + 2\pi m \quad (28)$$

same values of $\vec{E}$ field. With $\Delta x_{||} = \frac{2\pi m}{k}$ and $\lambda = \frac{2\pi}{k}$ the minimal non-zero $\Delta x_{||}$.

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{\omega} \frac{\omega}{k} = Tv \quad (29)$$

6.1.jpg

Figure 1:

where T is the period. ν frequency given by

$$\frac{1}{T} = \nu = \frac{\omega}{2\pi} \quad (30)$$

Going back to Maxwell

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad (31)$$

$$i\vec{k} \times \vec{E} = i\omega \vec{B} \quad (32)$$

$$\vec{\nabla} \cdot \vec{B} = 0 \implies \vec{k} \cdot \vec{B} = 0 \quad (33)$$

$$\vec{\nabla} \cdot \vec{E} = 0 \implies \vec{k} \cdot \vec{E} = 0 \quad (34)$$

$$\vec{E}, \vec{B} \perp \vec{k} \quad (35)$$

transverse wave.

$$\vec{E} \cdot (\vec{k} \times \vec{E}) = \vec{E} \cdot \vec{B} = 0 \quad (36)$$

where $\text{LHS} = 0 \implies \vec{E} \perp \vec{B}$. Using right hand rule we can find perpendicular